

RESEARCH ARTICLE

Ecosystem function in predator–prey food webs—confronting dynamic models with empirical data

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Abstract

1. Most ecosystem functions and related services involve species interactions across trophic levels, for example, pollination and biological pest control. Despite this, our understanding of ecosystem function in multitrophic communities is poor, and research has been limited to either manipulation in small communities or statistical descriptions in larger ones.
2. Recent advances in food web ecology may allow us to overcome the trade-off between mechanistic insight and ecological realism. Molecular tools now simplify the detection of feeding interactions, and trait-based approaches allow the application of dynamic food web models to real ecosystems. We performed the first test of an allometric food web model's ability to replicate temporally nonaggregated abundance data from the field and to provide mechanistic insight into the function of predation.
3. We aimed to reproduce and explore the drivers of the population dynamics of the aphid herbivore *Rhopalosiphum padi* observed in ten Swedish barley fields. We used a dynamic food web model, taking observed interactions and abundances of predators and alternative prey as input data, allowing us to examine the role of predation in aphid population control. The inverse problem methods were used for simultaneous model fit optimization and model parameterization.
4. The model captured >70% of the variation in aphid abundance in five of ten fields, supporting the model-embodied hypothesis that body size can be an important determinant of predation in the arthropod community. We further demonstrate how in-depth model analysis can disentangle the likely drivers of function, such as the community's abundance and trait composition. Analysing the variability in model performance revealed knowledge gaps, such as the source of episodic aphid mortality, and general method development needs that, if addressed, would further increase model success and enable stronger inference about ecosystem function.

5. The results demonstrate that confronting dynamic food web models with abundance data from the field is a viable approach to evaluate ecological theory and to aid our understanding of function in real ecosystems. However, to realize the full potential of food web models, in ecosystem function research and beyond, trait-based parameterization must be refined and extended to include more traits than body size.

KEYWORDS

agricultural pests, allometry, body mass, conservation biological control, herbivore suppression, multitrophic functioning, predator–prey interactions, species traits

1 | INTRODUCTION

Human activity is affecting species and ecosystems world-wide with potentially far-reaching consequences for ecosystem functions and services (Foley et al., 2005; Young, McCauley, Galetti, & Dirzo, 2016). To predict the impacts from global change, we need to understand the drivers of ecosystem functions. From biodiversity–ecosystem function research, we know that plant biomass production increases with plant species richness and that sampling effects and niche complementarity are the two mechanisms driving this relationship (Cardinale et al., 2009; Duffy, 2002). Much less is known about the mechanisms underpinning functions involving multiple trophic levels, such as pollination and predation. In experiments with small predator–prey assemblages, positive biodiversity–ecosystem function relationships are most common, but neutral and negative relationships occur frequently (Griffin, Byrnes, & Cardinale, 2013; Letourneau, Jedlicka, Bothwell, & Moreno, 2009), likely driven by predator interference and intraguild predation (Cardinale et al., 2009, 2012; Ives, Cardinale, & Snyder, 2005). For larger communities, however, driving mechanisms remain a black box, as we are restricted to statistical descriptions relating community (trait) composition to multitrophic function (Albrecht, Duelli, Schmid, & Müller, 2007; Gagic et al., 2015; Tylanakis, Tscharnkte, & Klein, 2006). To improve our ability to predict function in real ecosystems, we need to understand how species interactions drive function in communities of realistic size and complexity.

Food web ecology has made significant progress in recent years, and dynamic food web models have been suggested as a means to overcome the trade-off between community complexity and mechanistic insight into multitrophic ecosystem function (Cardinale et al., 2012). These models describe the temporal abundance dynamics of consumer and resource populations, accounting for the effect of trophic interactions. By default, the models also map several ecosystem functions, such as standing stock, production and consumption at multiple trophic levels. These models are increasingly used to address how biodiversity and trophic complexity affect ecosystem function in theoretical communities, providing well-founded expectations on the role of these factors in real ecosystems (e.g., Allhoff & Drossel, 2016; Poisot, Mouquet, & Gravel, 2013; Schneider, Brose,

Rall, & Guill, 2016; Wang & Brose, 2018). Though less common, dynamic food web models are also applied to describe the abundance dynamics or trophic interactions of real predator–prey assemblages (Boit, Martinez, Williams, & Gaedke, 2012; Jonsson, Kaartinen, Jonsson, & Bommarco, 2018; Schneider, Scheu, & Brose, 2012). This is a potentially powerful approach for evaluating ecological theory against empirical data and for understanding the function of real ecosystems. The power of the approach comes from the fact that dynamic food web models intrinsically embody hypotheses about the mechanisms underlying the functional outcome. For example, a dynamic food web model describing predation rate as a function of predator–prey trait matching is expected to replicate observed population fluctuations well, but only if trait matching is actually a determinant of predation in the real community. Therefore, confronting dynamic food web models with observations of predator–prey communities could be an important link between theory and empirical evidence and provide an opportunity to gain mechanistic insights into multitrophic functioning.

Two main obstacles to this approach have been the difficulty of establishing the food web structure, that is, determining who eats whom, and of estimating the values for the often numerous model parameters. But recent advances might allow us to overcome these obstacles. Firstly, novel DNA-based tools for gut content analysis have reduced the effort of establishing food web structure compared with more traditional techniques (Furlong, 2015; Traugott, Kamenova, Ruess, Seeber, & Plantegenest, 2013). DNA-based tools also expand the range of communities for which food web structure can be established. A case in point is arthropod communities, which typically include animals of small size and cryptic habits, a prevalence of soft-bodied, quickly disintegrating prey and predators with chewing or sucking mouthparts (Traugott et al., 2013). Molecular gut content analysis now allows us to establish food web structure in these ubiquitous and functionally important communities (e.g., Roubinet et al., 2017).

Secondly, trait-based approaches show promise to predict species interactions (Bartomeus et al., 2016; Eklöf et al., 2013). In particular, food web model parameterization has become easier thanks to allometric (body size based) approaches (Boit et al., 2012; Yodzis, 1998). Previously, parameterization required every trophic link to

be independently estimated, prohibiting the use of dynamic food web models to map the abundance dynamics of large predator–prey assemblages. In contrast, the allometric approach assumes a general relationship between organismal body size and metabolism (Brown, Gillooly, Allen, Savage, & West, 2004a; Brown, Gillooly, Allen, Savage, & West, 2004b) and from this infers the relationship between body size and trophic interaction strength (Schneider et al., 2012). The allometric functions needed for model parameterization include normalization constants that are currently not well known; they must still be estimated from empirical data for any given community. Nevertheless, by assuming that these allometric constants are the same across taxa, only a handful of constants need to be estimated (Rall, Kalinkat, Ott, Vucic-Pestic, & Brose, 2011; Schneider et al., 2012). Despite these simplifying assumptions, there are inspiring examples of allometric food web models realistically mapping typical seasonal multitrophic dynamics in a large lake (Boit et al., 2012) and trophic interaction strengths in microcosm experiments (Jonsson et al., 2018; Schneider et al., 2012). Thus, the approach of confronting dynamic food web models with empirical abundance or interaction data is now feasible and potentially very powerful.

The few previous applications of this approach have been successful, but the data used were either generalized population abundances averaged across a decade (Boit et al., 2012) or interaction strengths in the controlled environment of experimental microcosms (Jonsson et al., 2018; Schneider et al., 2012). It remains to be tested whether the approach can be useful when a dynamic food web model is confronted with temporally nonaggregated field data, or whether the inevitable noisiness of such data will undermine the power of the approach to provide insight into multitrophic function in naturally occurring communities. To test this, we confronted an allometric dynamic food web model with data from a naturally occurring, terrestrial arthropod predator–prey community. Specifically, we aimed to replicate the population dynamics of an herbivorous pest species, the bird cherry–oat aphid (*Rhopalosiphum padi*), in ten Swedish barley fields. The dynamic food web model embodied the hypotheses that predator and prey body mass and abundance determine the function of predation and that the balance between predation mortality and temperature-dependent intrinsic aphid population growth determines the level of the related service of aphid population regulation. The model took as input data the food web structure, derived from molecular analysis of field-collected predator individuals, body mass estimates for all taxa in the food web, and field derived abundance time series for the predators and their alternative prey; the model output was aphid abundance dynamics. We estimated the allometric constants for the model by minimizing the difference between the model-generated and observed aphid abundance time series. We used an “inverse problem” approach to parameter estimation, which optimizes the fit of a model to the data it describes (Banks, Hu, & Thompson, 2014; Tarantola, 2005). This flexible method for model specification is a common approach in diverse fields, including hydrogeology (Carrera, Alcolea, Medina, Hidalgo, & Slooten, 2005), epidemiology (Banks, Davidian, Samuels, & Sutton, 2009), ecotoxicology (Banks, Dick, Banks, & Stark, 2008) and ecology (Adams, Banks,

Banks, & Stark, 2005). We found that the model successfully captured the aphid dynamics in five of ten fields, providing support for the model-embodied hypotheses. We show how an in-depth analysis of a successful model can reveal the likely drivers of predation rates and observed herbivore population dynamics, such as changes in the abundance and body size composition of the predator community. In addition, we show how our approach can reveal knowledge gaps—both system-specific and general—that need to be addressed to further increase model success and the ability of the approach to provide robust insights into ecosystem function.

2 | MATERIALS AND METHODS

2.1 | Population density estimates

Abundances of the bird cherry–oat aphid, *Rhopalosiphum padi*, its main predators, spiders, carabid and coccinellid beetles, and their alternative prey, thrips, dipterans, collembolans and earthworms, were sampled in 10 spring barley fields in the province of Uppland, Sweden. Sampling occurred during 6 weeks from late May to early July 2011, corresponding to the period the bird cherry–oat aphid is damaging the crop. Temperatures were recorded in each field every 15 min using temperature loggers (TK-0014 Tinytag Tinytalk 2, Gemini Data Loggers).

Different sampling techniques were used for aphids, other plant-dwelling arthropods, soil organisms and ground-dwelling predators. Within each field, aphids were counted on 100 barley tillers approximately once a week throughout the study period. Similarly, thrips, dipterans and coccinellids were collected in 50 sweepnet sweeps, 2–4 times during the study period. Soil samplers of Ø5 and Ø25 cm were used to sample collembolans and earthworms, respectively, twice during the study period. Carabids and spiders were collected in six wet pitfall traps per field. They were emptied approximately once a week and left open in the interim; see Roubinet et al. (2017) for further details on sampling.

For use in the model, population densities were estimated [individuals/m²] by dividing abundance by the area sampled (see Banks et al., 2016). Thus, we estimated the area occupied by 100 barley tillers and the area swept by 50 sweeps and calculated the area of the soil samplers. However, the actual area being sampled is difficult to estimate for passive techniques, such as pitfall traps. Moreover, pitfall traps have a body mass bias (Arneberg & Andersen, 2003); large species have higher mobility than small species and therefore a higher likelihood to be caught in the traps (Spence & Niemelä, 1994). To mitigate this, we applied a bias-correcting method to the pitfall trap data, based on the expectation that species abundances at the local scale follow a negative three-quarter relationship with species body mass (Arneberg & Andersen, 2003) (see Supporting Information Appendix). After we had obtained the bias-corrected abundance, we estimated density for each field and sampling occasion by calculating the average daily catch and assuming a daily sampling area of $\pi \cdot 1.5^2 \text{ m}^2$ (Zhao et al., 2013).

For computational reasons, that is, avoiding zero densities, aphid density time series were shifted by adding the minimum detectable density: 4.1 [aphids/m²]. Furthermore, the time-series data were truncated; we focused on the part of the season during which the aphids are expected to exhibit exponential growth in the absence of predators. See Supporting Information Appendix for details.

2.2 | Food web structure

Each node in the food web corresponds to a taxon or group of taxa, and the links represent feeding interactions between these nodes. The food web was constructed using molecular gut content analysis (MGCA). Predators were collected in each field once a week during the periods 30 May – 12 June and 13 June – 26 June (see Roubinet et al., 2017). Individuals were subjected to whole-body DNA extraction and screened for DNA of predefined prey categories (the food web nodes) (see Staudacher, Jonsson, & Traugott, 2016). Thus, the taxonomic resolution of the food web was determined by the resolution of the prey categories used in the MGCA. For simplicity, we chose to interpret the category Aphididae as being constituted exclusively by *Rhopalosiphum padi*; this aphid species made up 98% of aphid abundance in the field.

To construct a binary food web detailing all potential trophic interactions between nodes, we pooled MGCA data from all fields and sampling occasions (Figure 1). However, to not give undue influence to very rare species, we excluded interactions observed for predator species with <10 analysed individuals. Information on feeding interactions for the predator nodes Tetragnathidae and Other

Araneae, that were too sparsely sampled, was based on the MGCA data of the ecologically similar taxa Linyphiidae and Lycosidae, respectively. Similarly, we added feeding interactions that were expected, based on the literature, but not detectable using MGCA, for example, cannibalism (see Supporting Information Appendix).

2.3 | Body masses

If the taxonomic resolution in the abundance data was the same as in the food web (for nodes Aphididae, Thysanoptera and Lumbricidae), the node was assigned a field-independent body mass. However, if the resolution in the abundance data was higher than in the food web (for Collembola, Diptera and all predator nodes), the assigned body mass was field specific, depending on the species composition in each field, and was calculated as:

$$W_f = \sum_{i=1}^{t_f} w_i \frac{1}{S_f} \sum_{j=1}^{s_f} \frac{n_{i,j,f}}{\sum_{k=1}^{t_f} n_{k,j,f}} \quad (1)$$

That is, the mass W_f for a node with t_f constituting taxa and s_f abundance samples in field f , is an average of the constituting taxa's body mass w_i , weighted by their across-season average of relative contribution to node density ($n_{i,j,f}$ is the population density of the i th taxa and at the j th sampling occasion in field f). The resulting node body masses are listed in Table 1. For a full description of body mass calculations and data, which were all taken from the literature, see Banks et al. (2016).

TABLE 1 Node body mass in milligram fresh weight. For nodes with field-specific body mass, the range across fields is given

Node Latin name	Node common name	Body mass (mg)
Aphididae	Aphids	0.6
Thysanoptera	Thrips	0.2
Diptera	Flies	26.7–29.7
Collembola	Springtails	1.5×10^{-3} – 2.1×10^{-3}
Lumbricidae	Earthworms	155.0
Bembidion	A genus of ground beetles	1.3–1.6
Harpalus	A genus of ground beetles	53.4–61.8
Poecilus	A genus of ground beetles	44.4–45.8
Pterostichus	A genus of ground beetles	85.2–104.0
Other Carabidae	Other ground beetles	2.3–21.3
Coccinella	A genus of ladybird beetles	4.5–20.8
Linyphiidae	Sheet weavers	1.0–1.6
Lycosidae	Wolf spiders	4.8–16.5
Tetragnathidae	Long-jawed orb weavers	1.9–2.3
Other Araneae	Other spiders	3.9–31.9

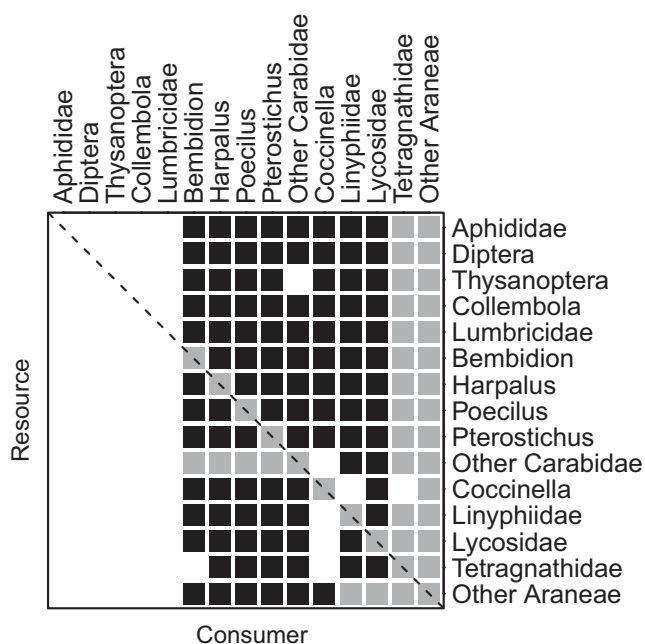


FIGURE 1 Food web structure used for all fields. Black squares are links based on molecular gut content analysis data from all fields and sampling times; grey links are from literature. Links on dashed line are cannibalistic. Nodes: $S = 15$, links: $L = 142$, connectance: $C = L/S^2 = 0.63$

2.4 | Dynamic food web model

To replicate the population dynamics of the aphid herbivore, we used a dynamic food web model (described below), as it allowed us to explicitly and mechanistically incorporate predation and the effect of predator abundance on aphid population growth. Essentially, what makes the model a food web model is not the number of species for whom the population dynamics are modelled, but that the population dynamics are an explicit function of trophic interactions. To compare predicted aphid population trajectories, with and without predation, we also used a model that did not take the food web interactions into account (i.e., Equation 2 without the last term; see Supporting Information eq. A4).

The aphid population density $N_{1,f}$ [ind/m²] in each field f was modelled using an allometrically parameterized dynamic food web model (Schneider et al., 2012):

$$\frac{dN_{1,f}}{dt} = N_{1,f}(t) \left[r(T_f(t)) - \sum_{j \in C_1} \frac{a_{1,j,f} N_{j,f}(t)}{1 + c_{j,f} N_{j,f} + \sum_{i \in R_j} a_{i,j,f} h_{i,j,f} N_{i,f}(t)} \right] \quad (2)$$

Here, $r(T_f)$ [day⁻¹] is the temperature-dependent growth rate of the aphid, T_f [°C] is the temperature in field f , $a_{i,j,f}$ [m² day⁻¹ ind _{j} ⁻¹] is the attack rate of predator j on prey i , $c_{j,f}$ [m² ind _{j} ⁻¹] is the coefficient of intraspecific interference competition of the predator, and $h_{i,j,f}$ [day ind _{j} ind _{i} ⁻¹] is the handling time for one individual of predator j to handle one individual of prey i . C_1 is the set of species that prey on the aphid, and R_j is the set of species that are prey to predator j . The model time unit t corresponds to one day. $N_{i,f}$ and $N_{j,f}$ [ind/m²], for all j and $i \neq 1$, are the population densities of the other (i.e., nonaphid) species in the food web. All parameters denoted by the subscript f are field specific.

The population densities of predators and alternative prey, that is, $N_{j,f}$ and $N_{i,f}$ [ind/m²] for all j and $i \neq 1$, were informed by the empirical data. The densities are given by linear functions in time, obtained from interpolations between empirical observations. At any time before the first observation or after the last observation, we linearly extrapolated from the two nearest data points (i.e., the first and second or second and second-from-last, respectively) (see Supporting Information Figures A3–A5 for the resulting community composition). Originally, we aimed to reproduce the observed abundance dynamics of all food web nodes. But the high and seemingly stochastic variability in the observed abundances of predators and alternative prey suggested that they were governed not only by the birth and death processes captured in the current model, but by many other factors, such as organism activity and phenology. The dynamic food web model would not be able to reproduce these effects without significant efforts in terms of model development, and more data and research to support this development. The observed aphid abundances, on the other hand, suggested that for the chosen time window predation and temperature-dependent growth could potentially be sufficient descriptors. Thus, we decided to focus on describing the aphid population dynamics and exploring its driving mechanisms.

The intrinsic growth rate, $r(T_f)$, of the aphid is a linear function of the temperature:

$$r(T_f) = 0.024T_f - 0.089 \quad (3)$$

This function is a linear regression on the observed rate of increase of *R. padi* kept at different temperatures (10–30°C) in the laboratory (Dean, 1974). The temperature T_f is a linear interpolation of daily average temperatures in each field.

The parameters $a_{i,j,f}$, $h_{i,j,f}$ and $c_{j,f}$ are body size dependent and were parameterized using the following allometric relationships (Schneider et al., 2012):

$$\begin{cases} a_{i,j,f} = a_{0,f} W_{i,f}^{0.25} W_{j,f}^{0.25} \left(\frac{W_{i,f}/W_{j,f}}{R_{opt,f}} e^{1 - \frac{W_{i,f}/W_{j,f}}{R_{opt,f}}} \right)^{\varphi_f} \\ h_{i,j,f} = h_{0,f} (W_{j,f}/W_{i,f})^{-0.25} \\ c_{j,f} = c_{0,f} W_{j,f}^{0.5} \end{cases} \quad (4-6)$$

Here, $W_{i,f}$ [mg ind _{i} ⁻¹] and $W_{j,f}$ [mg ind _{j} ⁻¹] are the body masses of prey i and predator j , respectively. $R_{opt,f}$ [ind _{i} ind _{j} ⁻¹] is the optimal predator–prey body mass ratio, for which attack success reaches the maximum value 1. $a_{0,f}$ [m² day⁻¹ mg^{-0.5} ind _{i} ^{0.25} ind _{j} ^{-0.75}], $h_{0,f}$ [day ind _{i} ^{-0.75} ind _{j} ^{-0.75}], and $c_{0,f}$ [m² mg^{-0.5} ind _{j} ^{-0.5}] are normalization constants; $a_{0,f}$, $h_{0,f}$ and $c_{0,f}$ gives the encounter rate, handling time and interference when $W_{i,f}^{0.25} W_{j,f}^{0.25}$, $(W_{j,f}/W_{i,f})^{-0.25}$, and $W_{j,f}^{0.5}$, respectively, equal unity. φ_f is a unitless width scaling parameter for the unimodal success rate curve, determining the sensitivity of the attack success to prey size. We estimated the values of the allometric parameters $a_{0,f}$, $h_{0,f}$, $c_{0,f}$, $R_{opt,f}$ and φ_f and the initial aphid density, $N_{1,0,f}$ (see below), from the empirical data.

2.5 | Model fit optimization and parameter estimation: the inverse problem

Predator and prey body mass are well known to be strong predictors of many types of trophic interactions, but the values of the allometric normalization constants (Equations 4–6) are not established, and must still be estimated from empirical data. In reality, these allometric constants might vary among systems (here fields) as well as among taxonomic or functional groups (here, e.g., consumer nodes). However, as there was not enough data to estimate allometric constants for each taxon in each field, we assumed allometric constants to be the same across taxa, but allowed them to vary across fields. This choice allowed us to optimize the model fit to the data from each field, which, in turn, should make it easier to discover whether there are important population regulation factors missing from the model. We thus estimated the values of the allometric normalization constants by solving an inverse problem for each field.

In an inverse problem, parameters are estimated by searching the parameter space for the parameter set that optimizes the fit between the mathematical model output and the empirical data (Banks et al., 2014). The fit is quantified by a cost function, which in turn is

defined by a statistical model. The formulation of the latter depends on the properties of the observation error.

Here, the mathematical model is the dynamic food web model, and its output was the modelled aphid population densities. Thus, we had the dynamical system in Equation 2 above, here represented as:

$$\frac{dN_{1,f}}{dt} = g(t, \mathbf{n}_f, \mathbf{q}_f) \quad (7)$$

Here, t is time, $\mathbf{n}_f$ is the vector of population densities of all nodes in field f , and $\mathbf{q}_f$ is the set of parameters to estimate. Specifically, $\mathbf{q}_f = [a_{0,f}, h_{0,f}, c_{0,f}, R_{opt,f}, \varphi_f, N_{1,0,f}]$, that is, the five constants of the allometric functions (Equations 4–6) and the initial aphid density, $N_{1,0,f}$, were estimated for each field independently. Estimating the initial condition, (here, $N_{1,0,f}$), instead of using the first empirical estimate, accounts for potential observation error and is particularly important if the model is sensitive to the initial condition.

The empirical data were the observed aphid population densities in each field (4–6 data points per field). We assumed an absolute error statistical model for the observations, which was supported by residual analysis (Banks et al., 2017). Thus, we assumed that the observations were of the form:

$$N_{1,j,f} = p(t_j, \mathbf{q}_{0,f}) + \varepsilon_{j,f} \quad (8)$$

where $N_{1,j,f}$ is the observed aphid population density at sampling occasion j in field f , the function p is the solution of the dynamical system in Equation 7 (and hence Equation 2), and $\varepsilon_{j,f}$ is the observation error. Thus, we assumed that there was some nominal (i.e., “true”) parameter set $\mathbf{q}_{0,f}$ given which the mathematical model, Equation 2, would describe the real aphid population dynamics in field f , which were observed with observation errors $\varepsilon_{j,f}$.

When assuming an absolute error statistical model, as we do here, we utilize the ordinary least squares (OLS) cost criterion (Banks & Tran, 2009; Banks et al., 2014) given by

$$J_{OLS,f}(\mathbf{q}_f) = \sum_{j=1}^{S_f} [N_{1,j,f} - p(t_j, \mathbf{q}_f)]^2 \quad (9)$$

where $\mathbf{q}_f$ is a possible set of parameters in the parameter space and, thus, an estimate of the nominal parameter set, for each field f . This quantity increases with the difference between observed and modelled aphid population densities, taking a square of the differences so as to penalize for both over- and underestimates of the observed population. Then, the estimate of the nominal parameter set for each field is given by:

$$\hat{\mathbf{q}}_{OLS,f} = \arg \min_{\mathbf{q}_f} \sum_{j=1}^{S_f} [N_{1,j,f} - p(t_j, \mathbf{q}_f)]^2 \quad (10)$$

The argmin function takes as input a criterion ($J_{OLS,f}$) and possible arguments (parameters $\mathbf{q}_f$), and it outputs the argument which minimizes the criterion. The estimate $\hat{\mathbf{q}}_{OLS,f}$ is therefore the mathematical argument (one of the possible inputs $\mathbf{q}_f$ to the function $J_{OLS,f}(\mathbf{q}_f)$), which minimizes the deviation between the model solution $p(t, \mathbf{q}_f)$

and the empirical aphid population density data $N_{1,j,f}$. In principle, this argument should be the nominal parameter set, $\mathbf{q}_{0,f}$, for which the only contributions to the cost criterion arise from the observational errors, $\varepsilon_{j,f}$, but in practice, scarcity of data and model complexity causes $\hat{\mathbf{q}}_{OLS,f} \neq \mathbf{q}_{0,f}$. For seeking the $\mathbf{q}_f$ which minimized the cost function, we used the minimizer function `fmincon` and ODE solver `ode23s` in MATLAB version 8.3.0.532 (R2014a).

2.6 | Model evaluation

We defined four criteria for assessing the performance of the dynamic food web model: (a) quantitative model fit, measured as R^2 , that is, the variation in observed aphid population density explained by the modelled population trajectory; (b) qualitative model fit, subjectively categorized as good, intermediate, or poor, based on visual investigation by 10 independent, unbiased observers of how well the model captured the presence and timing of population fluctuations (see Supporting Information Appendix for a full description); and (c) the ecological realism of the model parameter estimates; and (d) modelled process rates, as compared to direct empirical measurements published in the scientific literature.

3 | RESULTS

3.1 | Observed aphid dynamics

The aphid population dynamics differed substantially among fields (aphid population dynamics for each field are shown in the Supporting Information Figure A1). Based on the observed dynamics prior to the critical plant maturation stage, we distinguished four types of population trajectories: (a) strict increase (SI), (b) interrupted increase (II), (c) increase followed by late crash (ILC) and (d) bust-and-boom dynamics (BB) (Table 2). As the observed trajectory type influenced model performance, we present one example of each type (Figure 2). In SI fields, the aphid population density strictly increased over time (Figure 2a). In II-fields, the population increase was temporarily interrupted by a decline (Figure 2b), while in ILC fields the population experienced a large decline just before the critical maturation stage (Figure 2c). Finally, in BB fields, the aphid population fluctuated with high amplitude (Figure 2d).

3.2 | Quantitative and qualitative model fit

With regard to the first two model performance criteria, that is, quantitative and qualitative model fit, the result was variable. In the best case, for example, fields I, VI and IX, the dynamic behaviour was captured well and a large share (~90%) of the variation in the data was explained (Figure 2a, Table 2). In the worst cases (fields II and X), the model fit produced negative R^2 values, meaning that the mean of the data would have provided a better fit than the model (Figure 2d, Table 2). A good quantitative fit often mirrored a good qualitative fit, but there were exceptions. For example, in field III,

Field	Observed trajectory	Model trajectory	Qualitative fit	Quantitative fit (R^2)
V	Strict Increase	Exponential	Good	0.74
VI	Strict Increase	Exponential	Good	0.9
IX	Strict Increase	Exponential	Good	0.87
I	Interrupted Increase	Nonexponential	Good	0.88
IV	Interrupted Increase	Exponential	Intermediate	0.31
VII	Interrupted Increase	Nonexponential	Intermediate	0.26
VIII	Increase w Late Crash	Nonexponential	Good	0.09
X	Increase w Late Crash	Exponential	Poor	-0.97
II	Bust-and-Boom	Nonexponential	Poor	-0.5
III	Bust-and-Boom	Nonexponential	Intermediate	0.76

TABLE 2 Characterization of observed and modelled aphid population trajectories. Fields are categorized by observed trajectory type and, within each category, ordered by field number

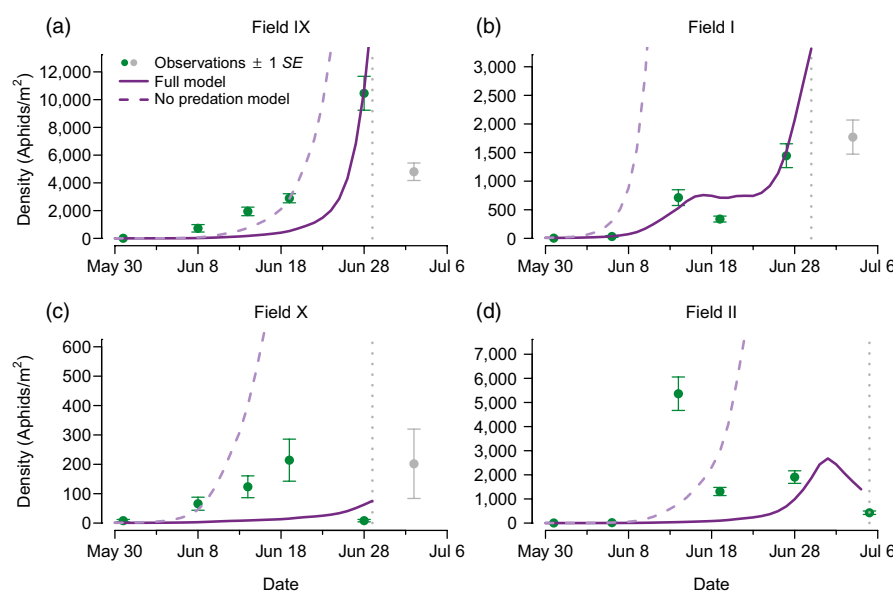


FIGURE 2 Observed and modelled aphid population densities over time in four fields with different types of observed trajectory: (a) strict increase, (b) interrupted increase, (c) increase with late crash and (d) bust-and-boom dynamics. Full model includes predation mortality and exponential aphid growth; no predation model includes only the latter. Vertical dashed line: the cut-off date, after which exponential aphid growth is unrealistic due to food quality decline. Observations in grey are from after this date and not used to inform the inverse problem

which had a bust-and-boom dynamics, the model did not reproduce the first peak and subsequent crash, but still explained 76% of the observed variation, as the model perfectly hit all data points but the peak (Supporting Information Figure A1c). This illustrates the necessity of both qualitative and quantitative model fit criteria to assess model performance.

The performance of the model was related to the type of aphid population trajectory observed in the field. Generally, the model performed best when the observed trajectory was strictly increasing (Table 2). In these cases, the model captured the exponential tendency of the population, but consistently underestimated all data points except the very last (Figure 2a and Supporting Information Figure A1). However, the exponential trajectories do not mean that aphid population growth was determined solely by the intrinsic population growth rate. Predation played a role also in these cases, as

can be seen by comparing the data and the result of the full model to that of a model without predation, with the latter providing a worse fit (Supporting Information eq. A4, Figure 2 and Supporting Information Figure A1).

In two non-SI fields, the model produced an exponential trajectory, not surprisingly resulting in unsatisfactory fits to the data (fields IV & X, Table 2, Figure 2c and Supporting Information Figure A1). The nonexponential trajectories, produced for the other non-SI fields, provided a good qualitative and/or quantitative model fit in some cases, but did not guarantee it (fields I, II, III, VII & VIII, Table 2, Figure 2b,d and Supporting Information Figure A1). The nonexponential model trajectories often followed a path between, rather than through, the data points; how this behaviour impacted the model fit depended on the nature of the observed dynamics. In the fields with interrupted increase, the

model trajectories smoothed over some of the observed fluctuations, but captured the overall tendency of the population (e.g., field I, Figure 2b). In the fields with more complex dynamics (the ILC and BB data trajectories), this behaviour sometimes caused the model to miss the peaks and/or the crashes completely (e.g., fields X & II, Figure 1c,d).

3.3 | Ecological realism of model parameter values and processes

The solutions to the inverse problems yielded reasonable estimates for the allometric constants and initial aphid density in each field (Table 3). The parameter estimates varied substantially across fields, but there were emergent patterns. The value of φ was, for most fields, so low ($<10^{-2}$) as to render most success rates equal to one, which means that R_{opt} had little effect in the model; the predators were equally successful in capturing prey regardless of either predator or prey body size. A low φ value also means that a predator population's contribution to total predation pressure exerted on a prey simply reflects the predator's contribution to total predator density (Figure 3). Furthermore, both h_0 and c_0 tended to have low values, resulting in low handling times and weak interference effects in the model. In practice, this reduced the functional response in four of the 10 fields to a pure type I response (Table 3), which means that the *per capita* interaction strengths are independent of both prey and predator densities (at approx. $h_0 < 10^{-3}$ and $c_0 < 10^{-1}$, respectively) (Figure 4a) and, at the population level, that the magnitude of predation pressure is simply proportional to the absolute predator density (Figure 3b). In another two fields, the functional response was type I with interference, making the *per capita* interaction strength decline with increasing predator density (Figure 3c), while in the remaining four fields, the functional response was of type II, in which *per capita* interaction strength declines with increasing prey density (Figure 3b). Thus, we find three different functional responses for the same study system. But as the analysis of model process realism indicates (see below), and as we discuss further in section *Interfield variability of allometric*

constants of the Discussion, this is unlikely to be a true reflection of the ecology of the system.

The allometric constants are not yet sufficiently established to enable a direct assessment of the ecological realism of their estimated values (but for previous estimates, see Jonsson et al., 2018; Rall et al., 2011; Schneider et al., 2012). Similarly, there is insufficient empirical data available to allow for a systematic comparison with our estimates of attack rates ($a_{ij,f}$) and handling times ($h_{ij,f}$) for particular species pairs and intraspecific interference ($c_{j,f}$) (Supporting Information Tables A2–A4). Therefore, the third model performance criterion, the ecological realism of estimated model parameter values, could not be evaluated for lack of empirical information.

To evaluate the fourth model performance criterion, the ecological realism of the modelled processes, it is possible to examine the consumption rates (i.e., the predators' realized functional response [$\text{ind}_{R.\text{padi}} \text{ day}^{-1} \text{ ind}_j^{-1}$]) produced by the estimated model parameters. For simplicity, we restrict the analysis here to the consumption of the focal species, *R. padi*, by a subset of its predators (Figure 4d–f and Supporting Information Figure A2). Individual predator consumption rates of aphids remained within ecologically reasonable bounds (i.e., at or below observed maximum consumption rates) as long as aphid densities were low. In fields with functional response (FR) type I (i.e., fields with low h_0 and c_0), modelled consumption rates soon became unrealistic as aphid densities increased (e.g., at ~ 750 aphids/m² ≈ 2 aphids/tiller, Figures 1b and 4d). When the success rate was dependent on the predator–prey body mass ratio (i.e., fields with $\varphi \geq 10^{-2}$), it was only for predators for which the aphid had nearly optimal prey size that the consumption rates became unrealistic, despite a FR type I (Figure 4f and Supporting Information Figure A1e). When predators had a FR type II, aphid densities could reach higher levels ($\sim 3,000$ aphids/m² ≈ 7 aphids/tiller, Figures 1a and 4e) before consumption rates became unrealistic, even then they remained more reasonable than with FR type I (compare Figure 4d,e). This is particularly noteworthy, as the a_0 , and hence the attack rates, was an order of magnitude larger for the fields

TABLE 3 Parameter estimates and the resulting type of functional response. Values <0.01 are presented as order of magnitude for legibility. Fields are ordered as in Table 1

Field	N_1^0	a_0	R_{opt}	φ	h_0	c_0	Functional response
V	0.2	0.10	163	1.76	10^{-7}	0.11	Type I with Interference
VI	0.2	0.05	323	10^{-6}	10^{-4}	10^{-3}	Type I
IX	1.0	2.20	500	0.11	0.01	0.01	Type II
I	9.8	0.17	12	0.01	10^{-6}	10^{-5}	Type I
IV	10.6	1.26	300	10^{-3}	0.14	0.02	Type II
VII	1.0	0.17	344	0.02	10^{-3}	10^{-9}	Weak type II
VIII	0.5	0.16	817	10^{-3}	10^{-5}	0.13	Type I with Interference
X	1.1	0.08	20	10^{-3}	10^{-6}	10^{-8}	Type I
II	1.0	0.14	11	0.27	10^{-5}	10^{-3}	Type I
III	1.5	0.09	150	10^{-3}	10^{-3}	10^{-3}	Weak type II

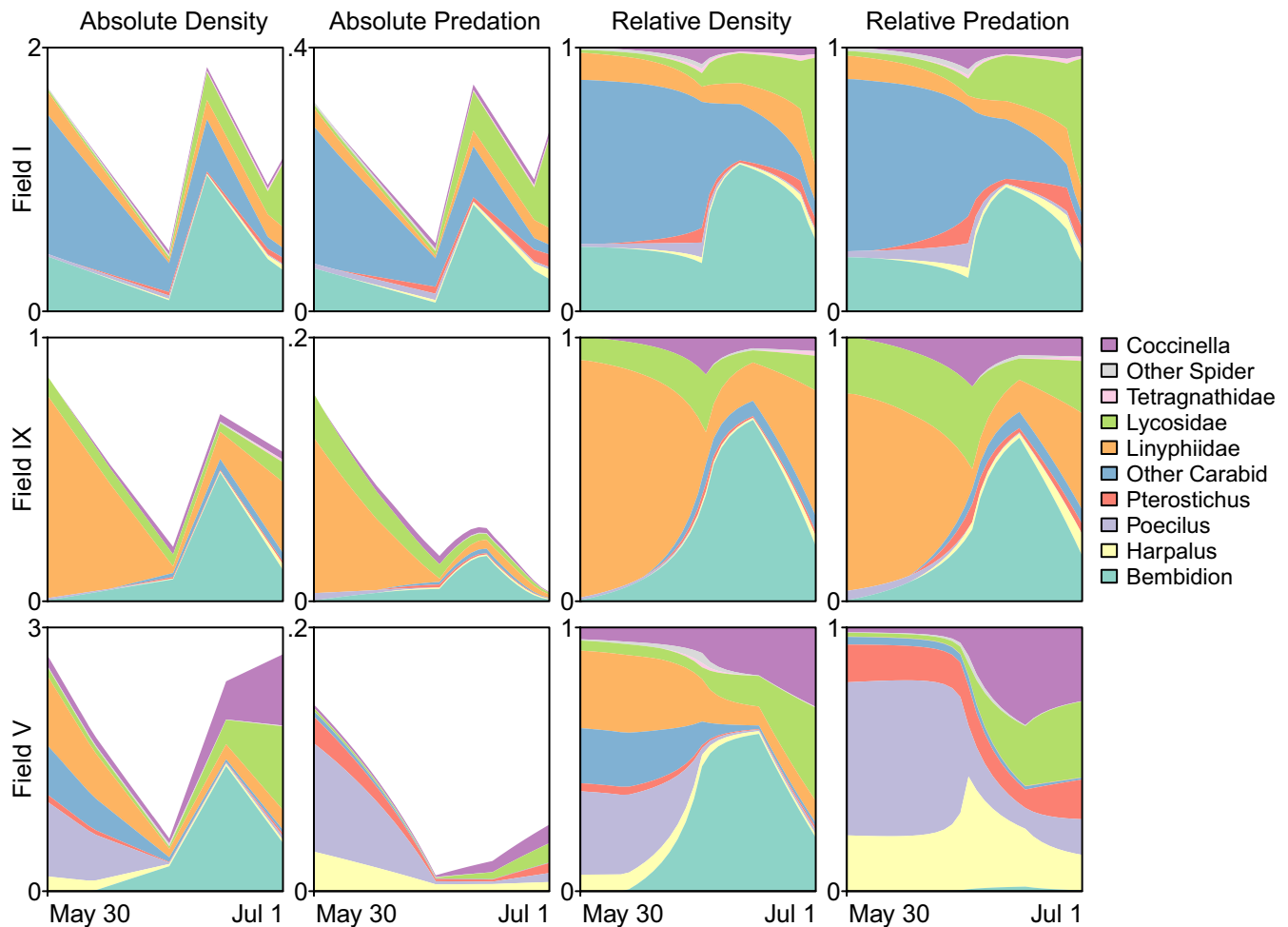


FIGURE 3 Absolute and relative predator densities and predation pressures on the aphids over time. Absolute density: predator individuals/m². Absolute predation: number of aphid individuals killed by the predator populations per day and aphid individual (i.e., the predation mortality term of Equation 2). Relative density: proportion of total predator density. Relative predation: proportion of total predation pressure on the aphid. Row 1: low ϕ and a type I functional response; absolute and relative predation pressure is proportional to the absolute and relative predator density, respectively. Row 2: Low ϕ and a functional response type II; absolute predation pressure reflects prey density and is hence not proportional to absolute predator density, while relative predation mirrors relative predator density due to the low ϕ . Row 3: High ϕ and predator interference; absolute and relative predation pressure is not proportional to the absolute and relative predator density, respectively. ϕ down-scales importance of predators for which the aphid is of nonoptimal size (e.g., Bembidion), and intraspecific interference dampens absolute predation pressure at high absolute density (e.g., Poecilus)

with FR type II than for the other fields (Table 3 and Supporting Information Table A2).

4 | DISCUSSION

4.1 | Insights into ecosystem function

The allometric food web model captured both the timing and magnitude of the aphid population dynamics, when the observed population trajectory was strictly increasing (three of 10 fields). In these cases, though the population trajectory remained exponential in shape, predation dampened the growth of the aphid population. For these, and two other fields with more complex dynamics, the food web model captured >70% of the variation in aphid abundance. The explanatory power is surprisingly large, considering that we

attempted to explain the population dynamics of an herbivore in the field, with a model including only temperature-dependent population growth and body size based predation mortality. To a large extent, this supports the model-embodied hypotheses that the body sizes and abundances of predators and prey determine the function of predation in this generalist arthropod community and that the balance between the predation and the temperature-dependent population growth determine the level of the related service of aphid population regulation.

When modelling a specific ecological system, extracting information from a well-performing model can inform us about the ecological processes operating in the real predator-prey community. Model validation against independent data would increase certainty of the model analysis, but in our case the limited amount of available data prohibited this. The model for the aphid population dynamics in

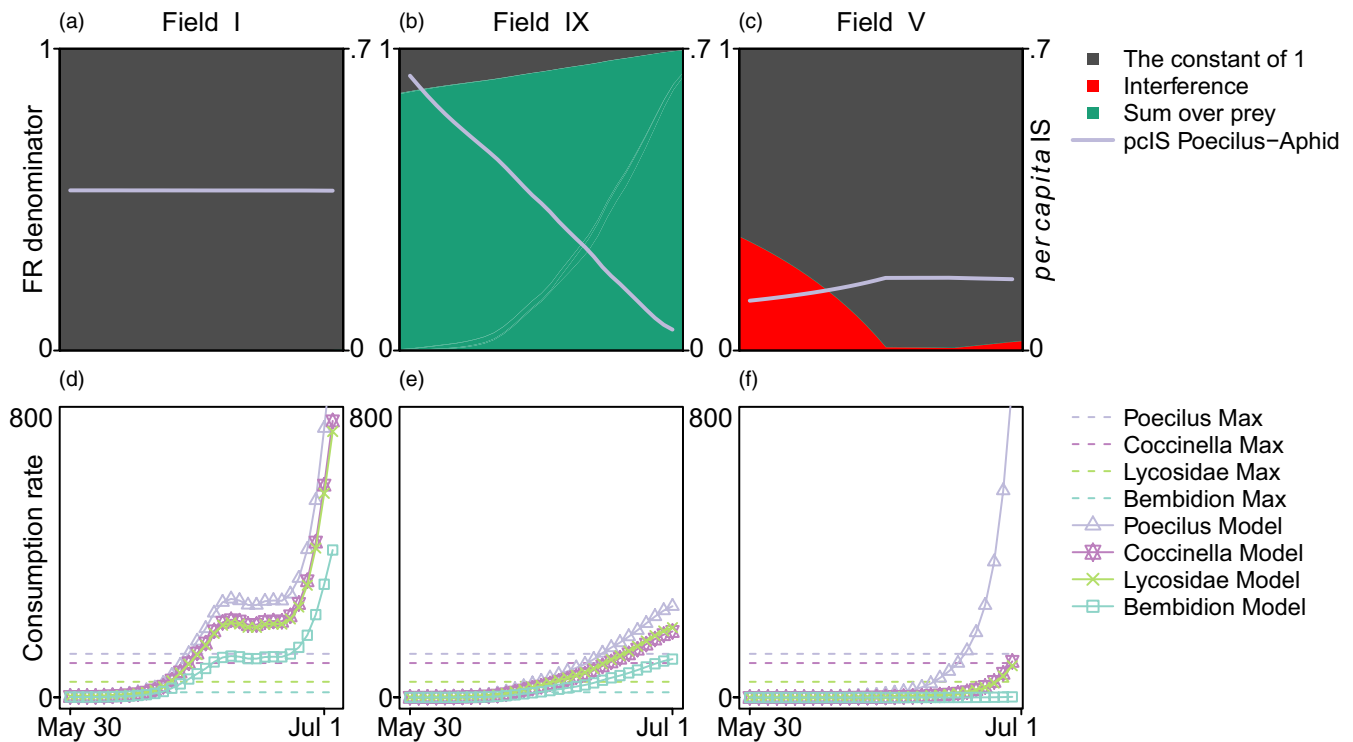


FIGURE 4 Aspects of trophic interaction strength over time. Row 1, left y-axis: *per capita* interaction strength (pclS) [$\text{m}^2 \text{day}^{-1} \text{ind}_{\text{Poecilus}}^{-1}$]. Row 1, right y-axis: surface plot showing the proportional contribution to the functional response denominator by its constituting components, for node Poecilus consuming aphids. (a) Low h_0 and c_0 values cause the constant 1 to dominate the functional response denominator, effectively reducing the functional response to type I. (b) High h_0 and low c_0 values cause the prey component ($\sum a_{ij}h_{ij}N_j$) to dominate, producing a functional response type II without interference effect; lower pclS at higher prey density. (c) Low h_0 and high c_0 cause the interference term to dominate, producing a functional response type I with interference effect; low pclS at higher predator density. Row 2: Consumption rates in the model and empirical maximum consumption rate estimates [Poecilus: 125, Coccinella: 99, Bembidion: 15 (Freier, Moeves, Triltsch, & Rappaport, 1998), Lycosidae: 45 (Mansour & Heimbach, 1993)]

field IX best met the full set of performance criteria. We therefore focus on this field as an example of how the model can be mined for information on ecosystem functioning. In field IX, the risk of predation for an aphid individual decreased over time (Figure 3: field IX, column 2). This loss of biological control mirrored the decline in total predator density, but while the predator abundance returned to its previous levels, the total predation pressure did not (Figure 3: field IX, columns 1–2). When predation is strongly governed by body size, a shift in the predator community from optimal to suboptimal sized predators can explain a disconnect between total predator density and predation pressure (e.g., Figure 3: field V, columns 1–2). But this disconnect had other causes in field IX; although larger predators (e.g., Harpalus and Lycosidae) contributed more to the total predation pressure than expected based on their relative abundance, the effect was small and the predator community was dominated by suboptimally sized predators (i.e., Linyphiidae and Bembidion) throughout the season (Figure 3: field IX, columns 3–4). Instead, it was the decrease in *per capita* interaction strength, driven by the growing aphid population (Figure 4b), that caused the disconnect between predator density and aphid predation risk. Thus, the graphic analysis of the model (Figures 3 and 4) revealed that, in field IX, the mid-season decline in predator abundance allowed the aphid population to quickly reach a size, at which the impact of predation

on aphid growth became negligible, due to the limitation on consumption posed by prey handling times. Confronting an allometric food web model with data here provided support for the hypotheses embodied by the model and the example from field IX illustrates how detailed insight into ecosystem function can be gained through the analysis of successful models. These results are encouraging as they show that field data, despite its inherent noisiness, can be used for confronting dynamic food web models, to test ecological theory and aid our understanding of ecosystem function.

4.2 | Knowledge gaps revealed

For the other five fields, the model was less successful in capturing aphid dynamics, indicating that in these fields there were important drivers of aphid dynamics in addition to the body size based predation and temperature-dependent population growth accounted for by the model. Though an in-depth analysis of less successful models cannot provide the insights into function as illustrated above, a similar in-depth investigation of the causes of noncompliance to the performance criteria can be informative. In particular, we identified system-specific knowledge gaps and model development needs, as well as general methodological developments that are needed to further integrate empirical and theoretical ecosystem function

research. Addressing these knowledge gaps and development needs would further increase model success and enable stronger inference about ecosystem function.

Poor model fits were due to the model not being able to fully replicate observed fluctuations, especially population crashes. We concluded that the cause of the population declines was more likely mortality than emigration, as few winged aphids were observed in the field. Among potential mortality factors, we ruled out insecticides, as farmers had been instructed to withhold application. We also ruled out predation from considered predators, as the model did not manage to recreate the aphid crashes, despite its capacity to recreate predation driven declines [for instance, late season aphid decline in field II (Figure 2d) was driven by increased coccinellid predation (Supporting Information Figure A6)]. Although unidentified, this mortality factor is dynamically important. Indeed, we have demonstrated elsewhere that including in the model an unspecified catastrophic mortality event prior to a population crash substantially improved model fit (Banks et al., 2017). Thus, the poor model fit revealed a knowledge gap, the source of a sudden strong aphid mortality, which must be addressed if we want to predict aphid population development.

Investigating process rates in the model revealed further knowledge gaps on aphid population regulation. We found that the predator *per capita* consumption rate of aphids was unrealistically high and, thus, overestimated in many fields (Supporting Information Figure A2). It is possible that the aphid population growth rate that we used (Dean, 1974) was overestimated in the laboratory. If aphid population growth in the field (in the absence of predation) were lower than in the laboratory, for example, due to decreased plant quality as the crop matures (Leather & Dixon, 1981), the predation pressure would be overestimated in the model. In fact, the laboratory-derived growth rates predicted dynamics well early in the season, when plant quality was high, but less so later on (Banks et al., 2016), suggesting that considering information on temporal plant quality variation would help in capturing aphid dynamics throughout the season.

Finally, the process rate analysis underscored the need to move towards truly quantitative sampling of arthropods. The commonly used methods for sampling arthropod abundance, for example, with pitfall traps, are not adapted for estimating population densities (Spence & Niemelä, 1994). Underestimated predator densities would yield the unrealistic *per capita* consumption rates seen in the model, even if the overall (population level) consumption in the community were correctly estimated. With more reliable estimates of population densities, the potential for mechanistic modelling to inform ecosystem functioning research would be substantially increased.

4.3 | Interfield variability of allometric constants

Despite the process sparsity of the dynamic model, the interaction dense food web resulted in a total of 294 parameters per field to be determined for the feeding interactions alone: attack rate and handling time for each pairwise interaction and intraspecific

interference for each predator node. The allometric parameterization approach enabled us to model this complex community, by reducing the number of trophic interaction parameters to estimate from 294 to only the five normalization constants of the allometric functions (Equations 4–6). To achieve this drastic reduction, but still optimize the model fit for each field, the allometric constants were assumed to be constant across taxa but estimated for each field independently (see Materials and Methods). However, the variation in allometric constant estimates across fields was greater than expected, a result which has implications for both the empirical aspects of the approach and the underlying framework of allometric theory.

Part of the across-field variation likely stemmed from the previously discussed population regulation factors, as the optimization process would be expected to assign as much as possible of their impact to factors present in the model, that is, the predation. But another part of the variation was due to the tendency to model reduction, that is, the tendency of the optimization process to reduce one or several allometric constants (ϕ , c_0 , and/or h_0 , depending on field, see Table 3), to the point where their effect on trophic interaction strength was lost. Ecologically, a nullified ϕ would imply that success rate of trophic interactions is independent of the predator and prey body size ratio, while a nullified c_0 and h_0 would imply that *per capita* consumption rate is independent of predator and prey density, respectively. However, rather than reflecting the ecology of the arthropod community, the tendency to model reduction likely stemmed from the complexity of the inverse problem, as we demonstrate below for the reduction of the functional response.

In the fields, for which the estimated allometric constants for handling time and/or interference were close to zero, the original functional response formulation (type II, with predator interference) was effectively reduced to a simpler one (type I, with or without interference). Models with a reduced functional response could capture the aphid population dynamics as well as models retaining the original complex formulation (compare, e.g., fields VI and IX). Indeed, we show elsewhere that there was, in our case, no statistical justification of using the more complex model—at least not as far as model fit was concerned (Banks et al., 2017). However, when examining the predator *per capita* consumption rates, it became clear that the simpler functional responses could realistically represent the mechanics of predation only at the low end of observed aphid densities (Figure 4 and Supporting Information Figure A2). We concluded that, rather than reflecting the ecology in the community, the model reduction indicated that the posed inverse problem was too complex. The number of parameters to estimate was still too large relative to the number of aphid density observations (Banks et al., 2017), despite the drastic parameter reduction achieved by the allometric assumptions. We show elsewhere that to reduce this source of variation and to better estimate the allometric constants, a higher sampling frequency is needed (Banks et al., 2017). Additionally, the timing of sampling can be important, as data from different time points can vary in how much information they contribute to the parameter estimation. To optimize data collection for parameter estimation, with regard to both frequency and timing of

sampling, we recommend that pre-experimental model analysis is performed when designing the sampling protocol (see, e.g., Banks, Banks, Rinnovatore, & Jackson, 2015; Laubmeier et al., 2018). To keep sampling rates feasible, we also suggest reducing the number of parameters to be inferred statistically. Some of the allometric constants could be measured directly in a controlled setting, instead of being estimated from field data. Suitable candidates for this are the constants for success rate (R_{opt} and φ) and handling time (h_0), as these should be independent of habitat structure, in contrast to the constants for encounter rate (a_0) and interference (c_0).

Finally, the fields had different predator community compositions (Supporting Information Figure A3). This would have caused the allometric constants to vary across fields, if the assumption of their taxonomic independence was violated. It is commonly assumed that allometric relationships can be described by the same parameters for all organisms within large taxonomic groups (e.g., Boit et al., 2012; Schneider et al., 2012; Yodzis, 1998). This assumption can be traced back to the foundation of the allometric framework (Brose, 2010; Brown et al., 2004a; Brown et al., 2004b; Peters, 1986; Yodzis & Innes, 1992), which consists of body size relationships with metabolism, population densities, movement speeds, etc., observed on a large scale, spanning many magnitudes of order of body size and taxa from multiple kingdoms. Although these relationships are well documented, they leave a large amount of variation unexplained (Brown et al., 2004a; Brown et al., 2004b), and there is increasing evidence of variation in allometric constants across taxonomic and functional groups. For example, the optimal predator-prey body mass ratio (R_{opt}) can vary with both body size (Gilljam et al., 2011) and taxonomic group (Brose et al., 2008), and the opportunity for interference (c_j) and prey encounter (a_{ij}) should be greater for active than for sit-and-wait predators (Dell, Pawar, & Savage, 2014; Skalski & Gilliam, 2001). Furthermore, the optimal body mass ratio (R_{opt}), the sensitivity to deviations from it (φ), and prey handling time (h_{ij}) are likely to differ for predators with extra-oral and internal digestion (Cohen, 1995, 1998). What remains unknown, however, is precisely when the variation across taxonomic and functional groups necessitates group-specific estimates, and, conversely, when universal approximations of allometric constants are justified. In line with previous indications, our results suggest that when attempting to model body size dependent processes for specific organisms (Jonsson, 2017) and ecosystems (Schneider et al., 2012), the assumption of universal allometric constants is too simplistic.

4.4 | Remarks on the role of food web resolution

A substantial amount of field and molecular work went into constructing the, for generalist arthropod communities, highly resolved trophic interaction structure. If the aim were to predict the abundance of one or a few specific species, exploring the sensitivity of the abundance predictions to food web resolution would be recommended. A low sensitivity should indicate that less effort could be spent in characterizing the food web without predictive strength being lost. However, the explanatory power of approaches based on

the traits of species or individuals is likely to decrease, as is mechanistic insight into ecosystem function, the more individuals and species are aggregated into nodes spanning a large range of trait values. We used a high-resolution trophic interaction structure with two aims: to test whether a food web model could replicate field observations of arthropod abundance and whether the model subsequently could provide information on the ecosystem function of predation and herbivore population regulation. We believe that this approach will improve understanding and predictability of multitrophic function, and we suggest that a goal of future research should be to increase interaction structure resolution further for more accurate and informative predictions of functions driven by trophic interactions across landscapes and management gradients.

5 | CONCLUSIONS

We here set out to replicate field observations of the population dynamics of an arthropod herbivore, the bird cherry-oat aphid, with the ultimate aim to test whether the approach of confronting a dynamic food web model with field data could be useful for ecosystem function research. We found that the food web model captured a large part of the natural variation in herbivore abundance and that subsequent model analysis could provide detailed insight into predation and herbivore population regulation. The explanatory power of the relatively process sparse and generally formulated dynamic food web model is encouraging, considering the multitude of factors that influence abundance dynamics in real ecosystems. This demonstrates that the approach of confronting dynamic food web models with field data can be a powerful tool to test ecological theory and aid our understanding of ecosystem function.

Although the results suggest that ecosystem function can be explained with little system-specific detail, the poor model fits in some fields indicate that the addition of two system-specific factors—the effect of plant maturation on intrinsic aphid population growth rate and the source of episodic aphid mortality—would improve model performance. Maximizing system-specific realism is not the goal, but unless the model, to a satisfactory degree, can replicate the population dynamics observed, the model is unlikely to provide meaningful information about the ecosystem function. As with any evaluation of theory and models against system-specific data, the approach taken here should be expected to require multiple cycles of model and theory development, empirical investigations and/or data collection. In this iterative process, poor model fits are important sources of information on knowledge gaps and development needs to be addressed in order to achieve a successful model.

In testing a model against field data from a specific ecosystem, the insights we gain pertain to this ecosystem. As with any place-based research, we can gain general understanding of ecosystem function by confronting models with data from different ecosystems. Instead of being a problem, the idiosyncrasy of place allows us to identify the extent to which general theory is locally applicable, if and how the theory needs to be modified to fit the specific ecosystem,

and map out the context dependency of necessary modifications (Billick & Price, 2010). Generally applicable insights can also arise as part of the iterative process. In our case, the results highlighted (a) the need for feasible, quantitative sampling methods of arthropod abundance, (b) the value of pre-experimental model analysis to optimize the spatiotemporal sampling design and (c) uncertainties in and development needs for the underlying allometric framework. In particular, we need to further explore and establish the allometric relationships between trophic interaction strength and body size, but it is likely that we also need to expand the allometric framework to include other traits. For example, traits such as hunting mode (Pawar, Dell, & Savage, 2012), microhabitat use (Schmitz, 2007) and prey defences (Kalinowski & DeLong, 2016) are likely to affect species interaction strengths, and multiple traits and species' (a)biotic context probably interact to shape pairwise species interactions (Bartomeus et al., 2016; Eklöf et al., 2013; Poisot, Stouffer, & Gravel, 2014). Such developments will be fundamental for realizing the full potential of dynamic food web models, as tools for understanding and predicting ecosystem functioning.

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AUTHORS' CONTRIBUTIONS

A.C., H.T.B., J.E.B., T.J., R.B. and A.N.L. conceived the ideas and designed methodology; M.J. and M.T. provided the empirical data; A.C., H.T.B. and A.N.L. performed modelling and data analysis; A.C., J.E.B., T.J. and R.B. led the writing of the manuscript. All authors contributed to the interpretation of results and writing and gave final approval for publication.

DATA ACCESSIBILITY

The MATLAB code and the data that was used for solving the inverse problems in this article are available from the Dryad Digital Repository: <https://doi.org/10.5061/dryad.41b5b06> (Curtsdotter et al., 2018).

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